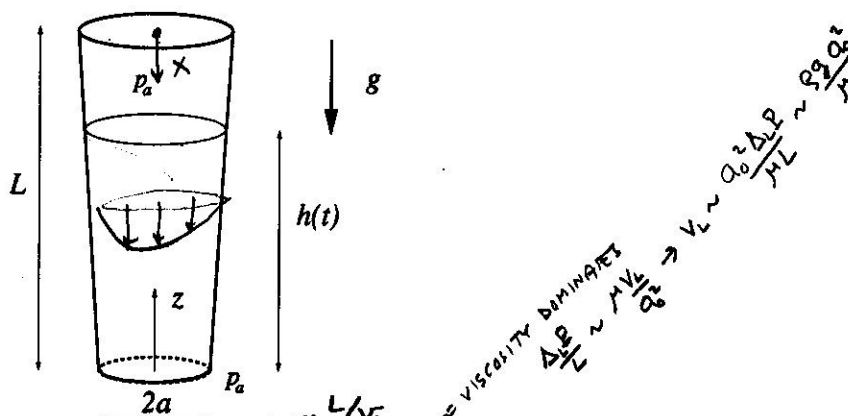


VF7

The vertical conical pipe of length L shown in the figure has a radius that increases linearly from the lower end according to $a(z) = a_0(1 + \alpha z/L)$, with $a_0 \ll 1$ and $\alpha \sim O(1)$. The pipe, open at the top, is initially full of a liquid of viscosity μ and density ρ . At a given time, the lower end opens up and the liquid discharges to the atmosphere under the action of gravity. Give the condition for the motion to be dominated by viscosity and, in that case, obtain the evolution of $h(t)$ as well as the time needed to complete the discharge.



THE MOTION IS DOMINATED BY VISCOSITY IF

$$\frac{a_0^2/(\mu kg)}{t_0} \sim \frac{9 \mu L a_0 Q}{\rho} \sim \frac{9 \mu L a_0 Q}{\rho}$$

$$\frac{9 \mu L a_0^4}{\rho^2 g} < 1$$

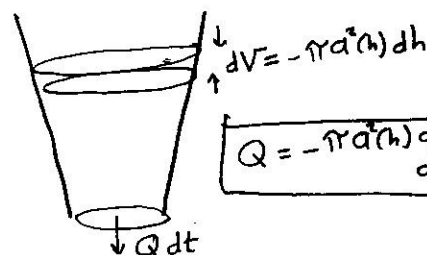
$$0 = -\frac{\partial P}{\partial x} + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) \Rightarrow v_x = \frac{P}{4\mu} (a^2 - r^2), \quad Q(t) = \frac{\pi}{8\mu} a^4 P$$

$$x = L - h, \quad P = P_a + \rho g h$$

$$\frac{\partial P}{\partial x} = -\frac{8\mu Q}{\pi a^4} \Rightarrow P_a + \rho g h - P = \frac{8\mu Q}{\pi} \int_{L-h}^L \frac{dx}{a^4} \Rightarrow \rho g h = \frac{8\mu Q}{\pi a_0^4} \int_{L-h}^L \frac{dx}{[1 + \frac{\alpha}{L}(L-x)]^4}$$

$$\rho g h = \frac{8\mu Q L}{\pi a_0^4} \left[1 - \frac{1}{(1 + \frac{\alpha h}{L})^3} \right]$$

TO CLOSE THE PROBLEM WE USE CONTINUITY TO WRITE



$$Q = -\pi a^2(h) \frac{dh}{dt}$$

$$\rho g h = -\frac{8\mu L}{\pi a_0^4} \left[1 - \frac{1}{(1 + \frac{\alpha h}{L})^3} \right] \frac{dh}{dt}$$

$$\frac{3\alpha}{8} \frac{\rho g a_0^2}{\mu L} \int_0^t dt = - \int_L^h \frac{(1 + \frac{\alpha h}{L})^3 - 1}{h(1 + \frac{\alpha h}{L})} dh$$

$h=0$ WHEN

$$t_d = \frac{8}{3\alpha} \frac{\mu L}{\rho g a_0^2} \int_0^L \frac{(1 + \frac{\alpha h}{L})^3 - 1}{h(1 + \frac{\alpha h}{L})} dh$$